Computational Hemodynamics and Physics-Guided Machine Learning for Arterial Stenosis Prediction

Jace E. Roman
College of Public Health, Temple University, Philadelphia, PA

Temple University
College of Public Health

Introduction

Arterial stenosis, the narrowing of a blood vessel, can substantially alter blood-flow velocity, pressure distribution, and wall shear stress. Computational fluid dynamics (CFD) enables controlled investigation of these changes but requires computationally intensive numerical simulations.

Machine-learning surrogate models may offer faster approximations of hemodynamic behavior. However, their reliability in predicting complex flow patterns, particularly downstream of severe narrowing, remains uncertain.

This research investigates how stenosis severity influences simulated hemodynamics and evaluates whether physics-guided machine learning improves axial blood-flow velocity predictions across idealized arterial geometries.

Materials and methods

Computational Modeling

Four idealized arterial geometries were constructed with a reference vessel diameter of 4 mm, a length of 120 mm, and nominal diameter reductions of 0%, 30%, 50%, and 70%. Three-dimensional blood-flow simulations were performed using SimVascular / svMultiPhysics. Blood was approximated as an incompressible Newtonian fluid with density 1,060 kg/m³ and dynamic viscosity 0.0035 Pa·s. Simulations used rigid, no-slip vessel walls, a prescribed steady inflow of 1,244 mm³/s, and a zero-traction outlet condition.

Machine Learning

A dataset of 132,740 CFD mesh-point samples from the 0%, 30%, and 50% geometries was used to evaluate axial-velocity predictions.
A geometry-only gradient-boosting regressor was compared against an analytical physics baseline and a physics-guided residual model. Leave-one-geometry-out evaluation tested performance on entire arterial geometries excluded from training. Peak velocity, pressure difference, wall shear stress, root mean square error (RMSE), and reverse-flow detection.

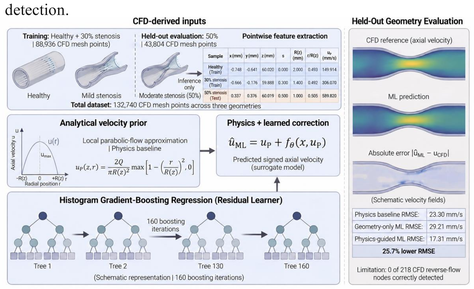


Fig. 1. Physics-guided histogram gradient-boosting pipeline for predicting arterial axial velocity. The model learns residual corrections to an analytical velocity estimate using CFD-derived features. Training used healthy and 30% stenosis geometries, with evaluation on a held-out 50% geometry. Architecture and flow-field illustrations are schematic.

Results

Hemodynamic Effects of Arterial Stenosis Results

Computational simulations demonstrated increasing peak blood-flow velocity, pressure loss, and wall shear stress with increasing arterial stenosis severity.
Peak velocity increased from 204.04 mm/s in the healthy artery to 786.77 mm/s in the 50% stenosis model, while maximum wall shear stress increased from 0.68 Pa to 5.73 Pa.
The 70% stenosis model produced a preliminary peak velocity of 2,119.00 mm/s, although downstream steady-state convergence was not established.

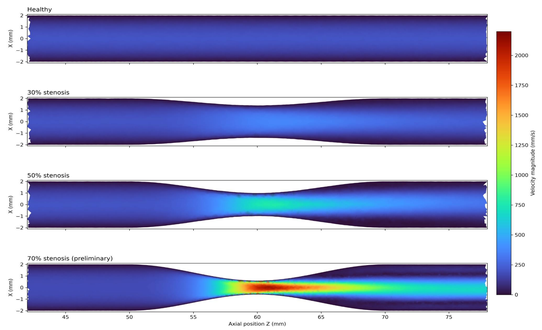


Fig. 2. Simulated blood-flow velocity distributions across four arterial geometries.

Machine-Learning Prediction Performance

A physics-guided gradient-boosting model trained on healthy and 30% stenosis geometries was evaluated against CFD results from an unseen 50% stenosis geometry.
The physics-guided model achieved an axial-velocity RMSE of 17.31 mm/s, compared with 23.30 mm/s for the analytical physics baseline, representing a 25.7% reduction in prediction error.
However, performance varied across geometry holdouts, demonstrating that improved prediction accuracy on one unseen geometry does not establish reliable generalization.

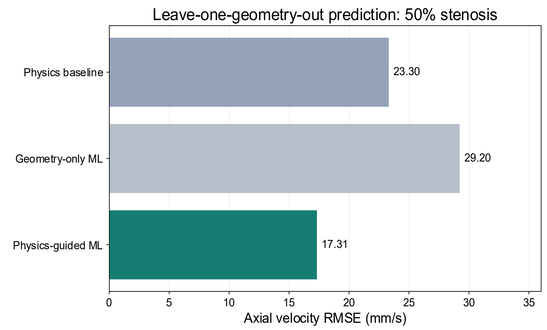


Fig. 3. Comparison of axial-velocity prediction errors between the analytical physics baseline, geometry-only machine learning, and physics-guided machine learning.

Wall Shear Stress Distribution

Maximum wall shear stress increased from 0.68 Pa in the healthy artery to 5.73 Pa at 50% stenosis. The 70% stenosis model exhibited a preliminary maximum WSS of 27.68 Pa, demonstrating substantial localized wall loading at the narrowed vessel region.

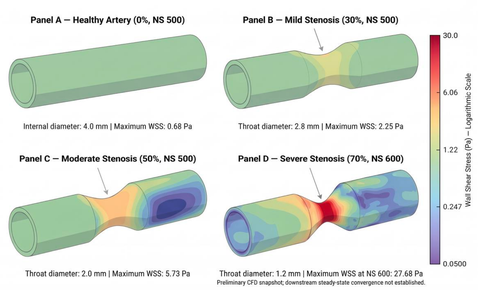


Fig. 4. Schematic wall shear stress distributions across four idealized arterial geometries. Maximum WSS measurements were obtained from CFD simulations. The 70% stenosis case represents NS 600 without demonstrated downstream steady-state convergence.

Reverse-Flow Prediction Limitations

Despite improved overall prediction accuracy, the physics-guided model failed to identify all 218 CFD nodes exhibiting reverse axial flow in the held-out 50% stenosis geometry.
These results indicate that reducing aggregate prediction error does not guarantee accurate reconstruction of localized hemodynamic disturbances.

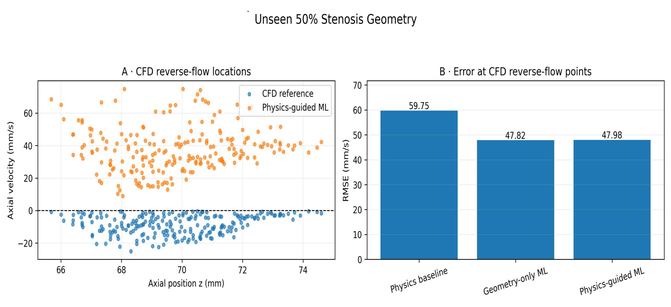


Fig. 5. Comparison of CFD reverse-flow observations and physics-guided ML predictions in the unseen 50% stenosis geometry.

Conclusions + Limitations

- Increasing stenosis severity increased simulated velocity, pressure loss, and wall shear stress.
- Physics-guided machine learning reduced axial-velocity RMSE on the held-out 50% stenosis geometry.
- The physics-guided model failed to capture reverse-flow behavior, while the 70% stenosis simulation did not demonstrate downstream steady-state convergence, despite an additional 100 Navier–Stokes solver steps (500 to 600). Consequently, the 70% stenosis results were classified as preliminary, requiring further numerical verification.

Literature cited

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Acknowledgments

Computational simulations were performed using SimVascular/svMultiPhysics. Data processing and machine-learning analyses utilized Python, NumPy, and scikit-learn. Research figures were developed using Python/Matplotlib in Visual Studio Code.

Figures 2, 3, and 5 present original computational results. Figures 1 and 4 were created with BioRender.com as scientific illustrations informed by the research methodology and CFD results.

Further information

Jace E. Roman
Temple University, College of Public Health
jace.roman@temple.edu

Scan for Research Code & Reproducibility

GitHub: github.com/z5rcnws7bz-eng/HemoDyn-Ai-v2